

CLASS X: CHAPTER - 8

INTRODUCTION TO TRIGONOMETRY

IMPORTANT FORMULAS & CONCEPTS

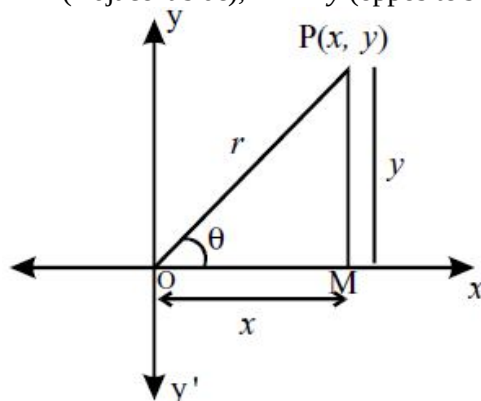
The word ‘trigonometry’ is derived from the Greek words ‘tri’ (meaning three), ‘gon’ (meaning sides) and ‘metron’ (meaning measure). In fact, **trigonometry** is the study of relationships between the sides and angles of a triangle.

Trigonometric Ratios (T - Ratios) of an acute angle of a right triangle

In XOY-plane, let a revolving line OP starting from OX, trace out $\angle XOP = \theta$.

From P (x, y) draw PM $\perp$ to OX.

In right angled triangle OMP. OM = x (Adjacent side); PM = y (opposite side); OP = r (hypotenuse).



$$\sin \theta = \frac{\text{Opposite side}}{\text{Hypotenuse}} = \frac{y}{r}$$

$$\operatorname{cosec} \theta = \frac{\text{Hypotenuse}}{\text{Opposite side}} = \frac{r}{y}$$

$$\cos \theta = \frac{\text{Adjacent Side}}{\text{Hypotenuse}} = \frac{x}{r}$$

$$\sec \theta = \frac{\text{Hypotenuse}}{\text{Adjacent Side}} = \frac{r}{x}$$

$$\tan \theta = \frac{\text{Opposite side}}{\text{Adjacent Side}} = \frac{y}{x}$$

$$\cot \theta = \frac{\text{Adjacent Side}}{\text{Opposite side}} = \frac{x}{y}$$

Reciprocal Relations

$$\sin \theta = \frac{1}{\operatorname{cosec} \theta}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

$$\cos \theta = \frac{1}{\sec \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{1}{\cot \theta}$$

$$\cot \theta = \frac{1}{\tan \theta}$$

Quotient Relations

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \quad \text{and} \quad \cot \theta = \frac{\cos \theta}{\sin \theta}$$

- **Remark 1 :** sin q is read as the “sine of angle q” and it should never be interpreted as the product of ‘sin’ and ‘q’
- **Remark 2 : Notation :** $(\sin \theta)^2$ is written as $\sin^2 \theta$ (read “sin square q”) Similarly $(\sin \theta)^n$ is written as $\sin^n \theta$ (read “sin nth power q”), n being a positive integer.
- **Note :** $(\sin \theta)^2$ should not be written as $\sin \theta^2$ or as $\sin^2 \theta^2$
- **Remark 3 :** Trigonometric ratios depend only on the value of θ and are independent of the lengths of the sides of the right angled triangle.

Trigonometric ratios of Complementary angles.

$$\sin(90 - \theta) = \cos \theta$$

$$\cos(90 - \theta) = \sin \theta$$

$$\tan(90 - \theta) = \cot \theta$$

$$\cot(90 - \theta) = \tan \theta$$

$$\sec(90 - \theta) = \operatorname{cosec} \theta$$

$$\operatorname{cosec}(90 - \theta) = \sec \theta.$$

Trigonometric ratios for angle of measure.

$0^\circ, 30^\circ, 45^\circ, 60^\circ$ and 90° in tabular form.

$\angle A$	0°	30°	45°	60°	90°
sinA	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cosA	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tanA	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not defined
cosecA	Not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
secA	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not defined
cotA	Not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

TRIGONOMETRIC IDENTITIES

An equation involving trigonometric ratios of an angle is said to be a trigonometric identity if it is satisfied for all values of θ for which the given trigonometric ratios are defined.

Identity (1) : $\sin^2\theta + \cos^2\theta = 1$

$$\Rightarrow \sin^2\theta = 1 - \cos^2\theta \text{ and } \cos^2\theta = 1 - \sin^2\theta.$$

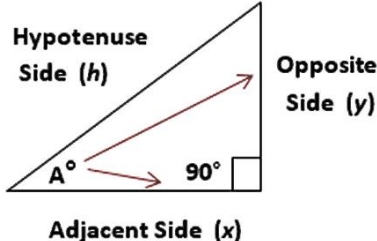
Identity (2) : $\sec^2\theta = 1 + \tan^2\theta$

$$\Rightarrow \sec^2\theta - \tan^2\theta = 1 \text{ and } \tan^2\theta = \sec^2\theta - 1.$$

Identity (3) : $\operatorname{cosec}^2\theta = 1 + \cot^2\theta$

$$\Rightarrow \operatorname{cosec}^2\theta - \cot^2\theta = 1 \text{ and } \cot^2\theta = \operatorname{cosec}^2\theta - 1.$$

SOME TIPS

Right Triangle	SOH-CAH-TOA Method	Coordinate System Method
 Hypotenuse Side (h) Opposite Side (y) Adjacent Side (x) A° 90°	SOH: $\sin(A) = \sin(A) = \frac{\text{Opposite}}{\text{Hypotenuse}}$ CAH: $\cos(A) = \cos(A) = \frac{\text{Adjacent}}{\text{Hypotenuse}}$ TOA: $\tan(A) = \tan(A) = \frac{\text{Opposite}}{\text{Adjacent}}$ $\operatorname{cosecant}(A) = \csc(A) = \frac{1}{\sin(A)} = \frac{\text{Hypotenuse}}{\text{Opposite}}$ $\secant(A) = \sec(A) = \frac{1}{\cos(A)} = \frac{\text{Hypotenuse}}{\text{Adjacent}}$ $\cotangent(A) = \cot(A) = \frac{1}{\tan(A)} = \frac{\text{Adjacent}}{\text{Opposite}}$	$\sin(A) = \frac{y}{h}$ $\cos(A) = \frac{x}{h}$ $\tan(A) = \frac{y}{x}$ $\csc(A) = \frac{1}{\sin(A)} = \frac{h}{y}$ $\sec(A) = \frac{1}{\cos(A)} = \frac{h}{x}$ $\cot(A) = \frac{1}{\tan(A)} = \frac{x}{y}$

Each trigonometric function in terms of the other five.

in terms of	$\sin \theta$	$\cos \theta$	$\tan \theta$	$\csc \theta$	$\sec \theta$	$\cot \theta$
$\sin \theta =$	$\sin \theta$	$\pm \sqrt{1 - \cos^2 \theta}$	$\pm \frac{\tan \theta}{\sqrt{1 + \tan^2 \theta}}$	$\frac{1}{\csc \theta}$	$\pm \frac{\sqrt{\sec^2 \theta - 1}}{\sec \theta}$	$\pm \frac{1}{\sqrt{1 + \cot^2 \theta}}$
$\cos \theta =$	$\pm \sqrt{1 - \sin^2 \theta}$	$\cos \theta$	$\pm \frac{1}{\sqrt{1 + \tan^2 \theta}}$	$\pm \frac{\sqrt{\csc^2 \theta - 1}}{\csc \theta}$	$\frac{1}{\sec \theta}$	$\pm \frac{\cot \theta}{\sqrt{1 + \cot^2 \theta}}$
$\tan \theta =$	$\pm \frac{\sin \theta}{\sqrt{1 - \sin^2 \theta}}$	$\pm \frac{\sqrt{1 - \cos^2 \theta}}{\cos \theta}$	$\tan \theta$	$\pm \frac{1}{\sqrt{\csc^2 \theta - 1}}$	$\pm \sqrt{\sec^2 \theta - 1}$	$\frac{1}{\cot \theta}$
$\csc \theta =$	$\frac{1}{\sin \theta}$	$\pm \frac{1}{\sqrt{1 - \cos^2 \theta}}$	$\pm \frac{\sqrt{1 + \tan^2 \theta}}{\tan \theta}$	$\csc \theta$	$\pm \frac{\sec \theta}{\sqrt{\sec^2 \theta - 1}}$	$\pm \sqrt{1 + \cot^2 \theta}$
$\sec \theta =$	$\pm \frac{1}{\sqrt{1 - \sin^2 \theta}}$	$\frac{1}{\cos \theta}$	$\pm \sqrt{1 + \tan^2 \theta}$	$\pm \frac{\csc \theta}{\sqrt{\csc^2 \theta - 1}}$	$\sec \theta$	$\pm \frac{\sqrt{1 + \cot^2 \theta}}{\cot \theta}$
$\cot \theta =$	$\pm \frac{\sqrt{1 - \sin^2 \theta}}{\sin \theta}$	$\pm \frac{\cos \theta}{\sqrt{1 - \cos^2 \theta}}$	$\frac{1}{\tan \theta}$	$\pm \sqrt{\csc^2 \theta - 1}$	$\pm \frac{1}{\sqrt{\sec^2 \theta - 1}}$	$\cot \theta$

Note: $\csc \theta$ is same as $\operatorname{cosec} \theta$.

MCQ WORKSHEET-I
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INTRODUCTION TO TRIGONOMETRY

1. In $\triangle OPQ$, right-angled at P, $OP = 7$ cm and $OQ - PQ = 1$ cm, then the values of $\sin Q$.
(a) $\frac{7}{25}$ (b) $\frac{24}{25}$ (c) 1 (d) none of the these
2. If $\sin A = \frac{24}{25}$, then the value of $\cos A$ is
(a) $\frac{7}{25}$ (b) $\frac{24}{25}$ (c) 1 (d) none of the these
3. In $\triangle ABC$, right-angled at B, $AB = 5$ cm and $\angle ACB = 30^\circ$ then the length of the side BC is
(a) $5\sqrt{3}$ (b) $2\sqrt{3}$ (c) 10 cm (d) none of these
4. In $\triangle ABC$, right-angled at B, $AB = 5$ cm and $\angle ACB = 30^\circ$ then the length of the side AC is
(a) $5\sqrt{3}$ (b) $2\sqrt{3}$ (c) 10 cm (d) none of these
5. The value of $\frac{2 \tan 30^\circ}{1 + \tan^2 30^\circ}$ is
(a) $\sin 60^\circ$ (b) $\cos 60^\circ$ (c) $\tan 60^\circ$ (d) $\sin 30^\circ$
6. The value of $\frac{1 - \tan^2 45^\circ}{1 + \tan^2 45^\circ}$ is
(a) $\tan 90^\circ$ (b) 1 (c) $\sin 45^\circ$ (d) 0
7. $\sin 2A = 2 \sin A$ is true when $A =$
(a) 0° (b) 30° (c) 45° (d) 60°
8. The value of $\frac{2 \tan 30^\circ}{1 - \tan^2 30^\circ}$ is
(a) $\sin 60^\circ$ (b) $\cos 60^\circ$ (c) $\tan 60^\circ$ (d) $\sin 30^\circ$
9. $9 \sec^2 A - 9 \tan^2 A =$
(a) 1 (b) 9 (c) 8 (d) 0
10. $(1 + \tan A + \sec A)(1 + \cot A - \operatorname{cosec} A) =$
(a) 0 (b) 1 (c) 2 (d) -1
11. $(\sec A + \tan A)(1 - \sin A) =$
(a) $\sec A$ (b) $\sin A$ (c) $\operatorname{cosec} A$ (d) $\cos A$
12. $\frac{1 + \tan^2 A}{1 + \cot^2 A} =$
(a) $\sec^2 A$ (b) -1 (c) $\cot^2 A$ (d) $\tan^2 A$

MCQ WORKSHEET-II
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1. If $\sin 3A = \cos (A - 26^\circ)$, where $3A$ is an acute angle, find the value of A .
(a) 29° (b) 30° (c) 26° (d) 36°
2. If $\tan 2A = \cot (A - 18^\circ)$, where $2A$ is an acute angle, find the value of A .
(a) 29° (b) 30° (c) 26° (d) none of these
3. If $\sec 4A = \operatorname{cosec} (A - 20^\circ)$, where $4A$ is an acute angle, find the value of A .
(a) 22° (b) 25° (c) 26° (d) none of these
4. The value of $\tan 48^\circ \tan 23^\circ \tan 42^\circ \tan 67^\circ$ is
(a) 1 (b) 9 (c) 8 (d) 0
5. If $\triangle ABC$ is right angled at C , then the value of $\cos(A + B)$ is
(a) 0 (b) 1 (c) $\frac{1}{2}$ (d) n.d.
6. The value of the expression $\left[\frac{\sin^2 22^\circ + \sin^2 68^\circ}{\cos^2 22^\circ + \cos^2 68^\circ} + \sin^2 63^\circ + \cos 63^\circ \sin 27^\circ \right]$ is
(a) 3 (b) 0 (c) 1 (d) 2
7. If $\cos A = \frac{24}{25}$, then the value of $\sin A$ is
(a) $\frac{7}{25}$ (b) $\frac{24}{25}$ (c) 1 (d) none of the these
8. If $\triangle ABC$ is right angled at B , then the value of $\cos(A + C)$ is
(a) 0 (b) 1 (c) $\frac{1}{2}$ (d) n.d.
9. If $\tan A = \frac{4}{3}$, then the value of $\cos A$ is
(a) $\frac{3}{5}$ (b) $\frac{4}{3}$ (c) 1 (d) none of the these
10. If $\triangle ABC$ is right angled at C , in which $AB = 29$ units, $BC = 21$ units and $\angle ABC = \alpha$. Determine the values of $\cos^2 \alpha + \sin^2 \alpha$ is
(a) 0 (b) 1 (c) $\frac{1}{2}$ (d) n.d.
11. In a right triangle ABC , right-angled at B , if $\tan A = 1$, then the value of $2 \sin A \cos A =$
(a) 0 (b) 1 (c) $\frac{1}{2}$ (d) n.d.
12. Given $15 \cot A = 8$, then $\sin A =$
(a) $\frac{3}{5}$ (b) $\frac{4}{3}$ (c) 1 (d) none of the these

MCQ WORKSHEET-III
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1. In a triangle PQR, right-angled at Q, $PR + QR = 25$ cm and $PQ = 5$ cm, then the value of $\sin P$ is
(a) $\frac{7}{25}$ (b) $\frac{24}{25}$ (c) 1 (d) none of these
2. In a triangle PQR, right-angled at Q, $PQ = 3$ cm and $PR = 6$ cm, then $\angle QPR =$
(a) 0° (b) 30° (c) 45° (d) 60°
3. If $\sin(A - B) = \frac{1}{2}$ and $\cos(A + B) = \frac{1}{2}$, then the value of A and B, respectively are
(a) 45° and 15° (b) 30° and 15° (c) 45° and 30° (d) none of these
4. If $\sin(A - B) = 1$ and $\cos(A + B) = 1$, then the value of A and B, respectively are
(a) 45° and 15° (b) 30° and 15° (c) 45° and 30° (d) none of these
5. If $\tan(A - B) = \frac{1}{\sqrt{3}}$ and $\tan(A + B) = \sqrt{3}$, then the value of A and B, respectively are
(a) 45° and 15° (b) 30° and 15° (c) 45° and 30° (d) none of these
6. If $\cos(A - B) = \frac{\sqrt{3}}{2}$ and $\sin(A + B) = 1$, then the value of A and B, respectively are
(a) 45° and 15° (b) 30° and 15° (c) 60° and 30° (d) none of these
7. The value of $2\cos^2 60^\circ + 3\sin^2 45^\circ - 3\sin^2 30^\circ + 2\cos^2 90^\circ$ is
(a) 1 (b) 5 (c) $5/4$ (d) none of these
8. $\sin 2A = 2 \sin A \cos A$ is true when A =
(a) 0° (b) 30° (c) 45° (d) any angle
9. $\sin A = \cos A$ is true when A =
(a) 0° (b) 30° (c) 45° (d) any angle
10. If $\sin A = \frac{1}{2}$, then the value of $3\cos A - 4\cos^3 A$ is
(a) 0 (b) 1 (c) $\frac{1}{2}$ (d) n.d.
11. If $3\cot A = 4$, then the value of $\cos^2 A - \sin^2 A$ is
(a) $\frac{3}{4}$ (b) $\frac{7}{25}$ (c) $\frac{1}{2}$ (d) $\frac{24}{25}$
12. If $3\tan A = 4$, then the value of $\frac{3\sin A + 2\cos A}{3\sin A - 2\cos A}$ is
(a) 1 (b) $\frac{7}{25}$ (c) 3 (d) $\frac{24}{25}$

MCQ WORKSHEET-IV
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1. Value of θ , for $\sin 2\theta = 1$, where $0^\circ < \theta < 90^\circ$ is:
(a) 30° (b) 60° (c) 45° (d) 135° .
 2. Value of $\sec^2 26^\circ - \cot^2 64^\circ$ is:
(a) 1 (b) -1 (c) 0 (d) 2
 3. Product $\tan 1^\circ \cdot \tan 2^\circ \cdot \tan 3^\circ \dots \tan 89^\circ$ is:
(a) 1 (b) -1 (c) 0 (d) 90
 4. $\sqrt{1 + \tan^2 \theta}$ is equal to:
(a) $\cot \theta$ (b) $\cos \theta$ (c) $\operatorname{cosec} \theta$ (d) $\sec \theta$
 5. If $A + B = 90^\circ$, $\cot B = \frac{3}{4}$ then $\tan A$ is equal to;
(a) $\frac{3}{4}$ (b) $\frac{4}{3}$ (c) $\frac{1}{4}$ (d) $\frac{1}{3}$
 6. Maximum value of $\frac{1}{\operatorname{cosec} \theta}$, $0^\circ < \theta < 90^\circ$ is:
(a) 1 (b) -1 (c) 2 (d) $\frac{1}{2}$
 7. If $\cos \theta = \frac{1}{2}$, $\sin \phi = \frac{1}{2}$ then value of $\theta + \phi$ is
(a) 30° (b) 60° (c) 90° (d) 120° .
 8. If $\sin(A + B) = 1 = \cos(A - B)$ then
(a) $A = B = 90^\circ$ (b) $A = B = 0^\circ$ (c) $A = B = 45^\circ$ (d) $A = 2B$
 9. The value of $\sin 60^\circ \cos 30^\circ - \cos 60^\circ \sin 30^\circ$ is
(a) 1 (b) -1 (c) 0 (d) none of these
 10. The value of $2\sin^2 30^\circ - 3\cos^2 45^\circ + \tan^2 60^\circ + 3\sin^2 90^\circ$ is
(a) 1 (b) 5 (c) 0 (d) none of these
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PRACTICE QUESTIONS
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INTRODUCTION TO TRIGONOMETRY
TRIGONOMETRIC RATIOS

1. If $\tan \theta = \frac{1}{\sqrt{5}}$, what is the value of $\frac{\operatorname{cosec}^2 \theta - \sec^2 \theta}{\operatorname{cosec}^2 \theta + \sec^2 \theta}$?
2. If $\sin \theta = \frac{4}{5}$, find the value of $\frac{\sin \theta \tan \theta - 1}{2 \tan^2 \theta}$.
3. If $\cos A = \frac{1}{2}$, find the value of $\frac{2 \sec A}{1 + \tan^2 A}$.
4. If $\sin \theta = \frac{\sqrt{3}}{2}$, find the value of all T-ratios of θ .
5. If $\cos \theta = \frac{7}{25}$, find the value of all T-ratios of θ .
6. If $\tan \theta = \frac{15}{8}$, find the value of all T-ratios of θ .
7. If $\cot \theta = 2$, find the value of all T-ratios of θ .
8. If $\operatorname{cosec} \theta = \sqrt{10}$, find the value of all T-ratios of θ .
9. If $\tan \theta = \frac{4}{3}$, show that $(\sin \theta + \cos \theta) = \frac{7}{5}$.
10. If $\sec \theta = \frac{5}{4}$, show that $\frac{(\sin \theta - 2 \cos \theta)}{(\tan \theta - \cot \theta)} = \frac{12}{7}$.
11. If $\tan \theta = \frac{1}{\sqrt{7}}$, show that $\frac{(\operatorname{cosec}^2 \theta - \sec^2 \theta)}{(\operatorname{cosec}^2 \theta + \sec^2 \theta)} = \frac{3}{4}$.
12. If $\operatorname{cosec} \theta = 2$, show that $\left\{ \cot \theta + \frac{\sin \theta}{1 + \cos \theta} \right\} = 2$.
13. If $\sec \theta = \frac{5}{4}$, verify that $\frac{\tan \theta}{(1 + \tan^2 \theta)} = \frac{\sin \theta}{\sec \theta}$.
14. If $\cos \theta = 0.6$, show that $(5 \sin \theta - 3 \tan \theta) = 0$.
15. In a triangle ACB, right-angled at C, in which AB = 29 units, BC = 21 units and $\angle ABC = \theta$. Determine the values of (i) $\cos^2 \theta + \sin^2 \theta$ (ii) $\cos^2 \theta - \sin^2 \theta$
16. In a triangle ABC, right-angled at B, in which AB = 12 cm and BC = 5 cm. Find the value of $\cos A$, $\operatorname{cosec} A$, $\cos C$ and $\operatorname{cosec} C$.
17. In a triangle ABC, $\angle B = 90^\circ$, AB = 24 cm and BC = 7 cm. Find (i) $\sin A$, $\cos A$ (ii) $\sin C$, $\cos C$.

PRACTICE QUESTIONS
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INTRODUCTION TO TRIGONOMETRY
T – RATIOS OF SOME PARTICULAR ANGLES

Evaluate each of the following:

1. $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ$

2. $\cos 60^\circ \cos 30^\circ - \sin 60^\circ \sin 30^\circ$

3. $\cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ$

4. $\sin 60^\circ \sin 45^\circ - \cos 60^\circ \cos 45^\circ$

5. $\frac{\sin 30^\circ}{\cos 45^\circ} + \frac{\cot 45^\circ}{\sec 60^\circ} - \frac{\sin 60^\circ}{\tan 45^\circ} - \frac{\cos 30^\circ}{\sin 90^\circ}$

6.
$$\frac{\tan^2 60^\circ + 4 \cos^2 45^\circ + 3 \operatorname{cosec}^2 60^\circ + 2 \cos^2 90^\circ}{2 \operatorname{cosec} 30^\circ + 3 \sec 60^\circ - \frac{7}{3} \cot^2 30^\circ}$$

7. $4(\sin^4 30^\circ + \cos^4 60^\circ) - 3(\cos^2 45^\circ - \sin^2 90^\circ) + 5 \cos^2 90^\circ$

8. $\frac{4}{\cot^2 30^\circ} + \frac{1}{\sin^2 30^\circ} - 2 \cos^2 45^\circ - \sin^2 0^\circ$

9. $\frac{1}{\cos^2 30^\circ} + \frac{1}{\sin^3 30^\circ} - \frac{1}{2} \tan^2 45^\circ - 8 \sin^2 90^\circ$

10. $\cot^2 30^\circ - 2 \cos^2 30^\circ - \frac{3}{4} \sec^2 45^\circ + \frac{1}{4} \operatorname{cosec}^2 30^\circ$

11. $(\sin^2 30^\circ + 4 \cot^2 45^\circ - \sec^2 60^\circ)(\operatorname{cosec}^2 45^\circ \sec^2 30^\circ)$

12. In right triangle ABC, $\angle B = 90^\circ$, AB = 3cm and AC = 6cm. Find $\angle C$ and $\angle A$.

13. If $A = 30^\circ$, verify that:

(i) $\sin 2A = \frac{2 \tan A}{1 + \tan^2 A}$ (ii) $\cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$ (iii) $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$

14. If $A = 45^\circ$, verify that

(i) $\sin 2A = 2 \sin A \cos A$ (ii) $\cos 2A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$

15. Using the formula, $\cos A = \sqrt{\frac{1 + \cos 2A}{2}}$, find the value of $\cos 30^\circ$, it being given that $\cos 60^\circ = \frac{1}{2}$

16. Using the formula, $\sin A = \sqrt{\frac{1 - \cos 2A}{2}}$, find the value of $\sin 30^\circ$, it being given that $\cos 60^\circ = \frac{1}{2}$

17. Using the formula, $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$, find the value of $\tan 60^\circ$, it being given that

$$\tan 30^\circ = \frac{1}{\sqrt{3}}.$$

18. If $\sin(A - B) = \frac{1}{2}$ and $\cos(A + B) = \frac{1}{2}$, then find the value of A and B.

19. If $\sin(A + B) = 1$ and $\cos(A - B) = 1$, then find the value of A and B.

20. If $\tan(A - B) = \frac{1}{\sqrt{3}}$ and $\tan(A + B) = \sqrt{3}$, then find the value of A and B.

21. If $\cos(A - B) = \frac{\sqrt{3}}{2}$ and $\sin(A + B) = 1$, then find the value of A and B.

22. If A and B are acute angles such that $\tan A = \frac{1}{3}$, $\tan B = \frac{1}{2}$ and $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$, show that $A + B = 45^\circ$.

23. If $A = B = 45^\circ$, verify that:

a) $\sin(A + B) = \sin A \cos B + \cos A \sin B$

b) $\sin(A - B) = \sin A \cos B - \cos A \sin B$

c) $\cos(A + B) = \cos A \cos B - \sin A \sin B$

d) $\cos(A - B) = \cos A \cos B + \sin A \sin B$

e) $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

f) $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

24. If $A = 60^\circ$ and $B = 30^\circ$, verify that:

a) $\sin(A + B) = \sin A \cos B + \cos A \sin B$

b) $\sin(A - B) = \sin A \cos B - \cos A \sin B$

c) $\cos(A + B) = \cos A \cos B - \sin A \sin B$

d) $\cos(A - B) = \cos A \cos B + \sin A \sin B$

e) $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

f) $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

25. Evaluate:

$$\frac{\sin^2 45^\circ + \frac{3}{4} \operatorname{cosec}^2 30^\circ - \cos 60^\circ + \tan^2 60^\circ}{\sin^2 30^\circ + \cos^2 60^\circ + \frac{1}{2} \sec^2 45^\circ}$$

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PRACTICE QUESTIONS
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T – RATIOS OF COMPLEMENTARY ANGLES

1. Evaluate: $\cot\theta \tan(90^\circ - \theta) - \sec(90^\circ - \theta) \operatorname{cosec}\theta + (\sin^2 25^\circ + \sin^2 65^\circ) + \sqrt{3} (\tan 5^\circ \cdot \tan 15^\circ \cdot \tan 30^\circ \cdot \tan 75^\circ \cdot \tan 85^\circ)$.
2. Evaluate without using tables: $\frac{\sec\theta \operatorname{cosec}(90^\circ - \theta) - \tan\theta \cot(90^\circ - \theta) + (\sin^2 35^\circ + \sin^2 55^\circ)}{\tan 10^\circ \tan 20^\circ \tan 45^\circ \tan 70^\circ \tan 80^\circ}$
3. Evaluate: $\frac{\sec^2 54^\circ - \cot^2 36^\circ}{\operatorname{cosec}^2 57^\circ - \tan^2 33^\circ} + 2 \sin^2 38^\circ \sec^2 52^\circ - \sin^2 45^\circ$.
4. Express $\sin 67^\circ + \cos 75^\circ$ in terms of trigonometric ratios of angles between 0° and 45° .
5. If $\sin 4A = \cos(A - 20^\circ)$, where A is an acute angle, find the value of A .
6. If A , B and C are the interior angles of triangle ABC , prove that $\tan\left(\frac{B+C}{2}\right) = \cot \frac{A}{2}$
7. If A , B , C are interior angles of a $\triangle ABC$, then show that $\cos\left(\frac{B+C}{2}\right) = \sin \frac{A}{2}$.
8. If A , B , C are interior angles of a $\triangle ABC$, then show that $\operatorname{cosec}\left(\frac{A+C}{2}\right) = \sec \frac{B}{2}$.
9. If A , B , C are interior angles of a $\triangle ABC$, then show that $\cot\left(\frac{B+A}{2}\right) = \tan \frac{C}{2}$.
10. Without using trigonometric tables, find the value of $\frac{\cos 70^\circ}{\sin 20^\circ} + \cos 57^\circ \operatorname{cosec} 33^\circ - 2 \cos 60^\circ$.
11. If $\sec 4A = \operatorname{cosec}(A - 20^\circ)$, where $4A$ is an acute angle, find the value of A .
12. If $\tan 2A = \cot(A - 40^\circ)$, where $2A$ is an acute angle, find the value of A .
13. Evaluate $\tan 10^\circ \tan 15^\circ \tan 75^\circ \tan 80^\circ$
14. Evaluate: $\left[\frac{\sin^2 22^\circ + \sin^2 68^\circ}{\cos^2 22^\circ + \cos^2 68^\circ} + \sin^2 63^\circ + \cos 63^\circ \sin 27^\circ \right]$
15. Express $\tan 60^\circ + \cos 46^\circ$ in terms of trigonometric ratios of angles between 0° and 45° .
16. Express $\sec 51^\circ + \operatorname{cosec} 25^\circ$ in terms of trigonometric ratios of angles between 0° and 45° .
17. Express $\cot 77^\circ + \sin 54^\circ$ in terms of trigonometric ratios of angles between 0° and 45° .
18. If $\tan 3A = \cot(3A - 60^\circ)$, where $3A$ is an acute angle, find the value of A .
19. If $\sin 2A = \cos(A + 36^\circ)$, where $2A$ is an acute angle, find the value of A .
20. If $\operatorname{cosec} A = \sec(A - 10^\circ)$, where A is an acute angle, find the value of A .



THANK YOU SO MUCH
BEST OF LUCK